

SUMMARY

Software Development Engineer with **1+ year of experience** building and optimizing **microservices**, **REST APIs**, and **SQL-driven applications** using **Java**, **Spring Boot**, and **SQL Server**. Backed by a strong undergraduate foundation in **computer science**, including **data structures**, **algorithms**, **databases**, **operating systems**, and **systems programming**, with hands-on experience in **compiler design**, **memory management**, and **performance optimization**.

EDUCATION

- Indraprastha Institute of Information Technology Delhi** Delhi, India
B.Tech in Computer Science and Design [2021 - 2025]

EXPERIENCE

- NatWest Group** Gurugram, Haryana, India
Software Development Engineer [July 2025 - Present]
 - Designed, developed, and maintained backend microservices using **Java**, **Spring Boot**, and **SQL Server**, following the Software Development Life Cycle (SDLC), **Agile** methodologies, **MVC** architecture, **Object-Oriented** Design principles, and **SOLID** design practices for enterprise-scale banking applications.
 - Engineered and enhanced **RESTful APIs** for service platforms, implementing **date-range filtering** capabilities for ticketing and audit-history services to improve data accessibility and user experience.
 - Optimized high-volume database operations by introducing **non-clustered indexing** and **query tuning strategies**, reducing ticket retrieval response times by approximately **50%**.
 - Built **API integrations** with **GitLab** and internal platform services to surface artifact ownership and team metadata, enabling governance, ownership visibility, and engineering asset discoverability.
 - Tech Stack:** Java, Spring Boot, SQL Server, REST APIs, Microservices, Git, GitLab, MVC Architecture, Object-Oriented Design, SOLID Principles, Agile Scrum

PROJECTS

- Custom Assembler and Simulator** Developed an assembler that translated assembly programs into machine code for a custom ISA and implemented a simulator to execute binary instructions. Built parsing, instruction decoding, register-state tracking, memory management, and control-flow execution modules, strengthening understanding of compiler pipelines and query-processing concepts.
- MeMS: Memory Management System** Built a memory management subsystem for a Real-Time Operating System using doubly linked lists for dynamic allocation and deallocation of memory blocks. Implemented memory utilization tracking and allocation algorithms to improve resource efficiency and runtime performance.
- NES Emulator** Implemented a 6502 CPU emulator supporting instruction decoding, execution, memory mapping, and hardware emulation for NES architecture. Designed modular system components and optimized execution performance for real-time gameplay.

SKILLS

- Languages:** C/C++(prof.), Java, Python, SQL, Javascript
- Backend Technologies:** Spring Boot, REST APIs, Microservices
- Databases:** SQL Server, MySQL, PostgreSQL
- Core Computer Science:** Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems
- Tools & Practices:** Git, GitLab, CI/CD, Agile Scrum, SDLC

LEADERSHIP EXPERIENCE

- Organizing Committee Member – Odyssey, IIIT Delhi** Coordinated logistics and execution of large-scale cultural events, collaborating with multiple teams and stakeholders to ensure timely delivery and smooth operations.
- Core Team Member – Media Services Committee, IIIT Delhi** Managed media coverage, content production, and live event operations for institute-level events, strengthening cross-functional collaboration and project coordination skills.
- Design & Social Media Head – Cultural Societies, IIIT Delhi** Led creative campaigns, content planning, and digital outreach initiatives, driving engagement while coordinating with student teams and event organizers.